Streptococcal Toxic Shock Syndrome

Streptococcal toxic shock syndrome (TSS) was first described in the late 1980s as a clinical entity similar to staphylococcal TSS but caused by invasive **group A streptococcus** (*Streptococcus pyogenes*). Recent evidence suggests that streptococcal TSS may be more common than the staphylococcal form.

The majority of cases are associated with **streptococcal pyrogenic exotoxin A (SPEA)**, although other superantigenic toxins such as **SPEB**, **SPEC**, and infections caused by **non–group A streptococci** have also been implicated.

Unlike staphylococcal TSS, streptococcal TSS is **not associated with tampon use** and can arise from nearly any type of group A streptococcal infection. The most frequent sources include **skin and soft tissue infections**, and it is well recognized as a complication of **varicella** and **influenza A**. However, in many cases, the precise portal of entry remains unidentified.

In contrast to staphylococcal TSS, **approximately 80% of streptococcal TSS cases originate from skin infections**. The initial presentation often includes **localized skin pain**, typically in an extremity. This pain progresses over several days to **localized erythema** and **edema**. The condition may rapidly evolve into **cellulitis**, **necrotizing fasciitis**, or **myositis**, with concurrent **bacteremia**.

A key diagnostic distinction is that **blood cultures are positive in more than 50% of streptococcal TSS cases**, compared to only about **10% in staphylococcal TSS**. Therefore, the presence of TSS symptoms combined with **localized cellulitis or soft tissue infection** should strongly suggest streptococcal TSS, as soft tissue involvement is uncommon in staphylococcal TSS.

Although individuals who are **very young, elderly, diabetic, or immunocompromised** are at increased risk, the majority of streptococcal TSS cases occur in **otherwise healthy adults**.

In Nonmenstrual Cases

In nonmenstrual cases—particularly those following surgery or associated with postoperative infections—the classic signs of localized infection such as **erythema, pain, and purulence** may be absent. This contrasts with the more typical presentation seen in streptococcal TSS, where soft tissue findings are prominent.